$\nu_\mu + \bar{\nu}_\mu$ charged-current production of two charged pions on argon in the NuMI beam

Selection, background constraint and extraction of the one-bin total (blind, pseudo-data)

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CC2$\pi^{\pm}Xp$ • FHC, RHC, combined • technical note draft 0.6

Outline

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To do and conclusions

Backup

Why two charged pions

- CC multi-pion production sits at the transition between resonant production (higher N^* and Δ states) and shallow and deep inelastic scattering, where models differ most.
- The observed charged-pion multiplicity is shaped by the intranuclear cascade: pions are absorbed, charge-exchanged or produced in re-scattering before they leave the nucleus, so a two-pion final state probes FSI directly.
- Multi-pion events are a background of the single-pion and pion-less measurements (a lost or misidentified pion); a measured rate on argon constrains the model used across the programme.
- NuMI at MicroBooNE: higher-energy tail than the BNB and 35–55% $\bar{\nu}_\mu$, so it complements the BNB CC2$\pi^{\pm}Np$ analysis.

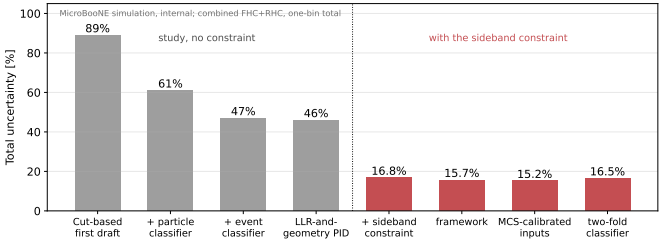
Goal of this stage

- A selection of $\nu_\mu + \bar{\nu}_\mu$ CC2$\pi^{\pm}Xp$ on the full NuMI exposure
- The one-bin total in FHC, RHC and combined, through the extraction framework, with a background constraint
- Validation on pseudo-data and checks of the inputs with beam-on data in the opened control regions
- Then the differential observables

The two-pion signal region is blind; every result shown is extracted from pseudo-data.

Starting point and history

- Built on the NuMI CC1$\pi^{\pm}$ analysis (Pophale, Nowak; release v1.9): samples, per-run POT normalisation, beam-off and dirt, systematic universes, detector variations, Wiener-SVD unfolding, MCS calibration of exiting tracks.
- First draft (2026-08-13; T. Mahmud, L. Gu): `CC1mu2pi`, `CC1mu3pi` as sub-selections of the single-pion selection with a looser pion BDT, Run-1 FHC overlay only. On the full exposure: 18.2% efficiency, 26% purity, expected total uncertainty 89%.
- Rebuilt since on the strategy of the BNB CC2$\pi^{\pm}$*Np* analysis (N. Majeed, internal note v2, May 2026) and on the current single-pion release.



Expected total uncertainty of the combined one-bin total along the analysis (the first four from the study outside the framework, without constraint).

Reused from the BNB CC2$\pi^{\pm}Np$ analysis

Element (BNB note)	In the NuMI analysis
Two-stage identification: per-track classifiers, best-assignment hypothesis, then an event classifier (BNB: purity 1% $\rightarrow$ 44%)	Adopted. The event classifier is where the purity comes from.
Assignment by the largest product of particle scores	Adopted: largest summed log-probability over all choices of one muon and N pions (at most eight tracks); best and runner-up for $N = 1, 2, 3$.
Event features: assigned scores, per-type sums, counts above threshold	Adopted, plus $L_2 - L_1$, $L_2 - L_3$, runner-up margin, track and shower counts, topological score, CosmicIP (33 features).
Background categories by topology	Adopted (CC1π, CC$\geq 3\pi$, CCπ^0, CC0π, NC, out of FV, dirt, beam-off).
Sidebands from the classifier outputs, each aimed at one migration path (1μ3π, 1μ3p, BDT band)	Adopted: three-pion-like, proton-like, score band, low score.
Conditional-covariance constraint (`ConstrainedCalculator`); BNB: 27.3% $\rightarrow$ 21.5%, not applied	Adopted after alternative-model closure: the lever on the $(1 + B/S)$ amplification of the flux.
One-bin total first, differential second; Recomb2 leads the detector term	Same order; Recomb2 also leads here.

Adapted for NuMI, and what is not reused

Adapted

- **Particle ID**: one four-class XGBoost (μ, π, p, other) instead of three binary classifiers; "other" = tracks without a truth match (cosmics). No NuGraph inputs (not in the NuMI ntuples). Deployed on LLR score and geometry only, for robustness against the detector variations.
- **Training**: no signal-enhanced NuMI samples; all 14 overlays. Particle classifier: whole run periods held out. Event classifier: two-fold by event parity, on the nominal mixture (overlay, dirt, beam-off, CV weights); the BNB meta-BDT saw enhanced samples only.
- **Signal**: $\nu_\mu + \bar{\nu}_\mu$, $p_\mu > 0.15$, every $p_\pi > 0.10\,\mathrm{GeV}/c$, π^0/kaon veto, any number of protons (BNB: ν_μ, KE > 10 MeV, $\geq 1p$). The $\geq 1p$ subsample (81%) is kept for comparison.
- **Working point**: on the expected total uncertainty of the one-bin total, not on efficiency $\times$ purity.
- **Containment**: not required (BNB: all π and p contained, a third of the efficiency).

Not reused

- KE > 10 MeV thresholds: that phase space is mostly not reconstructed.
- Invariant masses as results: the reco pion energy saturates, so the masses are squeezed towards threshold; study products unless a binning passes the 0.50 criterion.

Comparison

	BNB	NuMI (comb.)
Efficiency	4–6%	15%
Purity	47–58%	48%
Detector, predicted count*	11.0%	8.4%
Total, constrained	21.5%†	16.5%

* the seven BNB variations; † evaluated, not applied in the BNB note

Signal definition and scope

Signal: CC2$\pi^{\pm}Xp$

ν_μ or $\bar{\nu}_\mu$ CC, vertex in the fiducial volume of the single-pion analysis, one muon with $p_\mu > 0.15\,\text{GeV}/c$, exactly two charged pions leaving the nucleus, both with $p_\pi > 0.10\,\text{GeV}/c$ (a third charged pion at any momentum makes the event background); no π^0, kaon or heavier meson; any number of nucleons; $\theta(\mu, \text{leading } \pi) < 2.6$ rad.

- Leading = hardest pion (truth). Pion charge not measured.
- $0.10\,\text{GeV}/c$ is the pion tracking turn-on: it removes the unreconstructable 10% and keeps 90% of the two-pion signal. Efficiency is limited by reconstructing several tracks from one vertex, not by the soft pions.
- Events with a third charged pion below $0.10\,\text{GeV}/c$ amount to 6.2% of the signal (Run-4 FHC overlay) and are background by definition.
- Subsample with ≥ 1 proton above $0.3\,\text{GeV}/c$: 81%.

D1	Inclusive in nucleons, with a $\geq 1p$ subsample
D2	$p_\pi > 0.10\,\text{GeV}/c$; pion-momentum bins only above $0.175\,\text{GeV}/c$
D3	The two-pion signal region is blind
D4	NuGraph inputs: deferred (not available for NuMI)
D5	NuMI NuWro overlay: only if it can be validated
D6	Reco counterpart of the 2.6 rad opening-angle cut
D7	No exclusivity cut against the blind single-pion signal region
D8	Measurements stop at two pions; $\geq 3\pi$ is a control sample

Samples, exposure and blindness

- Samples and normalisation of the single-pion release: 14 ν_μ overlays (FHC Runs 1, 2, 4, 5; RHC Runs 1, 2, 3, 4a, 4b, 4c), 7 beam-off, 7 dirt samples, each run normalised to its own exposure.
- Exposure: 7.766×10^{20} POT FHC, 1.1082×10^{21} POT RHC, 1.8848×10^{21} POT combined.
- Detector variations: Run-4 FHC and RHC, CV + eight variations (LY down, LY Rayleigh, Recomb2, SCE, wire modification x, yz, angle xz, angle yz).
- Systematics: PPFX flux universes, GENIE multisim and unisims, Geant4 reinteraction, POT, targets, simulation statistics; detector variations taken fractionally (DVfrac).
- Pseudo-data: Poisson throws, run by run, of overlays and dirt at the data exposure; the framework adds the beam-off.
- Beam-on data used only in the opened control regions of the single-pion analysis, with both signal regions removed.



Starting point: the cut-based selection

	Gen. sig.	Sel. sig.	Sel. total	Eff.	Purity
2π FHC	1 210	221	836	18.2%	26.4%
2π RHC	1 672	303	1 168	18.1%	26.0%
2π comb.	2 882	524	2 004	18.2%	26.1%
3π comb.	551	54.5	517	9.9%	10.5%

- `CC1mu1piXp` preselection with a looser pion ID (vertex distance ≤ 9.5 cm, the first-draft pion BDT, up to two uncontained candidates), $\theta(\mu, \text{longest } \pi) < 2.6$ rad.
- Backgrounds: CC1π with a second candidate 22%, CCπ^0 21%, CC$\geq 3\pi$ 7%, NC 6%, beam-off 2.5%.
- Both candidates are true pions in only 60% of the selected signal.

Why purity is the lever

The flux uncertainty ($\approx 17\%$) scales signal and background together, so after background subtraction it enters the extracted total as

$$\delta_{\text{flux}} \approx 17\% \times (1 + B/S).$$

At 26% purity: flux 65%, cross section 47%, detector 34%, total 89%.

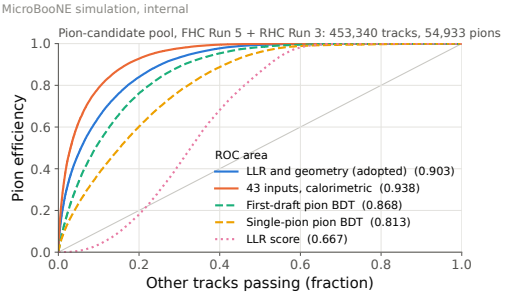
Two ways out: raise S/B (classifiers), and constrain B with control samples that share its composition (sidebands).

Particle classifier

- XGBoost, four classes (μ, π, p, other) on the tracks of the pion-candidate pool (generation 2, track score ≥ 0.3) of all 14 overlays: 3.1M tracks.
- Labels from the backtracked particle; tracks without a truth match (mostly overlay cosmics) are "other".
- Training: Runs 1, 2, 4 of both horn modes; held out: RHC Run 3, FHC Run 5.
- The Run 1–3 beam-on, beam-off and dirt ntuples lack 20 track branches, so the deployed models use the 43 inputs every sample carries.
- Evaluated in the selection through ROOT `RBDT`, identical to XGBoost.

Pion identification (63 inputs)	π/rest	π/p	π/μ
Particle classifier	0.954	0.985	0.895
First-draft pion BDT	0.888	0.945	0.804
Single-pion pion BDT	0.818	0.961	0.558
LLR score	0.670	0.893	0.251

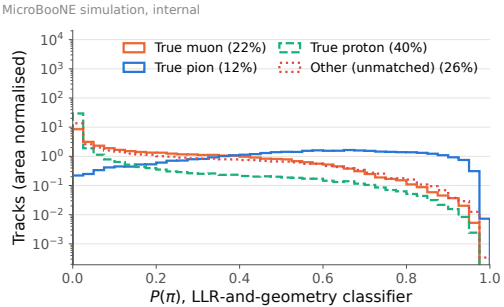
Table: held-out tracks with a truth match. Figure: the whole pool of the held-out runs, unmatched tracks as "other".



- At the first-draft pion efficiency (91.8%): candidate purity 27.6% $\rightarrow$ 43.3%, protons passing 13.4% $\rightarrow$ 4.1%.
- Same gain in FHC and RHC, for π^+ and π^-. Muon vs pion is the hardest separation.

The LLR-and-geometry particle classifier

- The detector variations move the calorimetric inputs (χ^2 scores, PIDA, Bragg likelihoods, calorimetric energies) by up to half their spread for protons; the LLR score by at most 0.04; geometric inputs not at all.
- The deployed classifier keeps only: LLR score (sum and per plane), length, track score, vertex distance, start and end containment, range and MCS momenta, proton-hypothesis energy, daughter counts, per-plane hit flags.
- Pion ROC area 0.892 vs 0.926 for the 43-input model over all held-out tracks (0.903 vs 0.938 on the pool); at 80% pion efficiency 9.0% of protons pass (vs 3.9%).
- It halves the part of the detector term that no constraint can reach (efficiency part 14.3% $\rightarrow$ 7.8%; next slides).



$P(\pi)$ of the deployed classifier by true class, held-out runs.

Track assignment

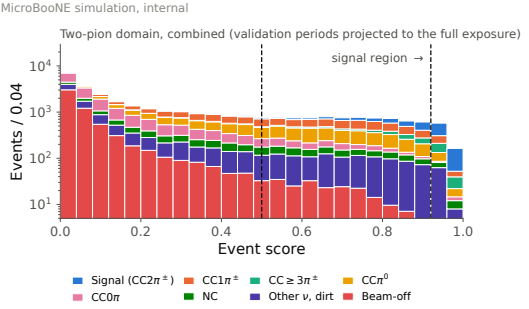
- Events with one neutrino slice and the vertex in the FV; tracks of the pool with track score ≥ 0.5, the muon candidate included, carry the four class probabilities.
- Hypothesis "one muon, N pions, the rest protons or other":

$$L_N = \log P_\mu(\mu) + \sum_{\text{pions}} \log P_\pi + \sum_{\text{rest}} \log\left(1 - P_\mu - P_\pi\right)$$

  probabilities floored at 10^{-6}; every choice among at most eight tracks enumerated (ranked by $\max(P_\mu, P_\pi)$); best and runner-up kept for $N = 1, 2, 3$.
- In the selected signal (held-out runs): the assigned muon is a true muon in 89%, each assigned pion a true charged pion in 88%, all three in 76%.
- In the preselected signal only 30% of events have the muon and both pions reconstructed as tracks, so in most signal events no assignment can be fully right: the event classifier has to work with partial topologies.
- Every track gets a role: second pion and proton candidates become available for observables, categories and the response.

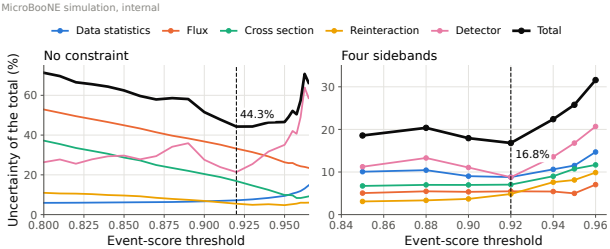
Event classifier

- XGBoost, two-pion signal vs everything else, 33 features: probabilities of the assigned tracks, $L_2 - L_1$, $L_2 - L_3$, margin over the runner-up, class sums and counts above 0.5, track and shower counts, topological score, CosmicIP.
- Domain: one slice, vertex in FV, software trigger, ≥ 3 tracks, $\theta(\mu, \text{longer assigned } \pi) < 2.6$ rad.
- Trained on Runs 1, 2, 4 (206k events, 13.6k signal) with the nominal mixture of overlay, dirt and beam-off and CV weights; tested on FHC Run 5 and RHC Run 3 (105k, 7.3k signal). The test periods also choose the working point (validation periods).
- ROC area 0.887. Kinematic features of the assigned tracks give 0.898, a gain that disappears in the total uncertainty: not used.
- Largest gains: summed $P(\pi)$, L_2, number of tracks with $P_\pi > 0.5$, topological score, number with $P_\mu > 0.5$.



Event score, test periods projected to the full exposure (combined).

Working point: minimum of the expected total uncertainty



- Threshold chosen on the expected total of the one-bin total with all terms: data statistics, flux, POT and targets, cross section, reinteraction, detector (eight Run-4 variations, FHC and RHC correlated), simulation statistics.
- Statistics and flux alone would push to 3.9% efficiency; the detector term rises steeply above 0.95. Minimum at 0.92, flat from 0.90 to 0.94.

Selection	Eff.	Pur.	Total
Cut-based `CC1mu2pi`	18.2%	26%	89%
+ particle classifier	15.8%	35%	61%
Event cl., 43 inputs	17.1%	56%	47%
Event cl., LLR and geometry	13.0%	51%	46%

Combined configuration, no constraint, score > 0.92. Flux 17% $\times (1 + B/S)$, except the last row (PPFX universes, as in the constraint study); the scan gives 44% at 0.92 with the proxy.

Selection requirements of CC1mu2piBDT

	Cut	Requirement
1	Slices	exactly one neutrino slice
2	Vertex	in the single-pion FV, outside the dead-wire region in z
3	Trigger	software trigger passed
4	Track score	≥ 0.5 (direct neutrino daughters)
5	PID inputs	LLR in $(-1, 2)$; p, μ, MIP Bragg in $(0, 500)$; length > 0
6	Ranking	by $\max(P_\mu, P_\pi)$; at most eight tracks
7	Pool size	at least three tracks
8	Assignment	muon and two pions maximising L_2
9	Opening angle	$\theta(\mu, \text{longer assigned } \pi) < 2.6$ rad
10	Event score	> 0.90 (two-fold classifier): signal region

- Cuts 1–9 define the domain of the event classifier; cut 10 the signal region.
- Cut 5 is a validity requirement of the pool (the Bragg likelihoods exist in every run; they are not classifier inputs).
- The single-pion cuts after its preselection (topological score > 0.67, muon, containment, wire-gap, shower and multiplicity cuts) are not applied; topological score, CosmicIP, track and shower counts enter only as features.
- Sidebands: cuts 1–9 and score ≤ 0.90 (next section).

The detector term

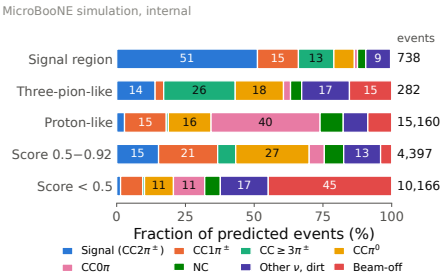
- With the 43-input classifier the detector term (33%) is the largest. A paired bootstrap (one Poisson weight per physical event, shared by CV and variations) attributes 7.7% of it to the statistics of the variation samples.
- Sources: Recomb2 (+22%), wire modification angle *yz* (+18%), *x* (+12%), *yz* (+8%); LY, SCE and angle *xz* consistent with zero.
- The variations lower the selected signal by $\approx 8\%$ and the background by 10–18%: an efficiency part, which no background constraint removes, and a background part.
- Training on the variation samples did not help. The LLR-and-geometry classifier halves the efficiency part and is adopted.
- In the BNB convention (predicted selected count): 10.6% (8 variations), 8.4% (the BNB seven), against 11.0% in the BNB note. The extracted total amplifies it by 2–4.

Particle cl.	Features	Pur.	Detector	(eff., bkg.)	Total
43 inputs	PID set	56%	32.7%	(14.3, 17.8)	47%
43 inputs	full set	62%	33.0%	(18.2, 14.8)	45%
no χ^2/PIDA/E	PID set	57%	35.1%	(14.9, 19.0)	48%
LLR+geom.	PID set	51%	21.4%	(7.8, 15.2)	44%
LLR+geom.	full set	55%	20.3%	(8.4, 12.9)	42%

Combined configuration, best threshold of each, no constraint. Totals within the 3% scatter of the scans.

Signal region and sidebands

- Four sidebands from the classifier outputs, exclusive, assigned in this order inside the two-pion domain (cuts 1–9, score ≤ 0.90):
  - three-pion-like: ≥ 4 tracks and $L_3 > L_2$ (the $\geq 3\pi$ control sample, D8; BNB $1\mu 3\pi$)
  - proton-like: an assigned pion with $P_p > 0.5$ (BNB $1\mu 3p$)
  - score band 0.5–0.90
  - low score < 0.5
- The sidebands leave out the events of the blind single-pion signal region, so they can be opened without it.
- Of the signal-region events 14% lie in the blind single-pion signal region (kept, D7) and 12% in its opened multi-π control region.



Predicted composition (single classifier), combined, full exposure; table: two-fold, all periods.

Region	Events	Signal	Purity
Signal region	920	444	48%
Three-pion-like	294	36	12%
Proton-like	16 640	458	3%
Score 0.5–0.90	4 534	655	14%
Score < 0.5	13 162	166	1%

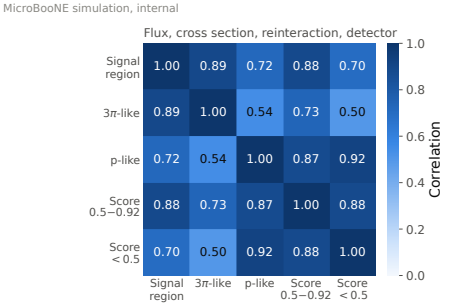
Background constraint: conditional covariance

- Covariance of the predicted counts in the signal region (s) and the sidebands (c), built from every source, conditioned on the sideband data d_c:

$$\hat{n}_s = n_s + V_{sc} V_{cc}^{-1} (d_c - n_c), \qquad V'_{ss} = V_{ss} - V_{sc} V_{cc}^{-1} V_{cs}$$

  (`ConstrainedCalculator`, the tool of the BNB note).
- The flux scales the background and the sidebands alike: flux 38% → 5.0%. Detector 23.5% → 9.3%, close to its efficiency part.
- Three-pion-like alone (294 events) halves the total. Data statistics grow, since the sideband counts enter the background estimate.

Constraint	Total	Flux	Xsec	Det.	Data st.
None	50.1%	38.2%	19.4%	23.5%	6.9%
Three-pion-like	27.8%	15.3%	9.5%	15.5%	11.9%
Proton-like	34.9%	13.3%	15.9%	26.3%	7.1%
Score 0.5–0.90	22.1%	5.1%	8.2%	17.3%	7.6%
Score < 0.5	35.5%	13.4%	10.9%	29.2%	7.5%
All sidebands	16.5%	5.0%	6.7%	9.3%	8.6%



Correlation of the predicted region counts under the systematic covariance (combined). Study outside the framework.

Closure under alternative models and the signal in the sidebands

Alternative-model closure

- Pseudo-data from a reweighted model in every region, extracted with the nominal model:
  - GENIE multisim universe 545: 0.965 (-0.21σ) constrained, 0.988 without
  - $\Delta \to N\pi$ angular: 0.995 (-0.03σ), 1.014 without
- Over all GENIE multisim universes: bias rms 4.0% with the constraint, every universe within 1σ; 20.3% without.
- These models lie inside the covariance: the test checks the procedure, not the model space.

Signal in the sidebands

- The score band holds more predicted signal than the signal region (655 vs 444). Signal variations move every region together, so the conditioning also carries the signal model from the sidebands into the signal region.
- Nominal (framework): the signal covariance between regions is kept, since the same flux and interaction parameters act coherently on one CC2π process in every region. The decorrelated variants are conservative stress tests:

Variant (study)	Constrained total
Nominal	16.5%
Sideband signal decoupled	19.6%
Without the score band	21.2%
Both	23.9%

Every variant closes within 0.25σ for both models. Score-band/SR signal ratio (single model): 1.684 nominal, 1.660 (u545), 1.681 ($\Delta \to N\pi$): these models do not probe the transfer; the stress model of the ensembles slides does.

Extraction through the framework

- `CC1mu2piBDT` in the selection framework: track pool, class probabilities (`RBDT`), assignment, 33 features, event score, region.
- On a 30k-event slice: features identical to the Python study, scores to 6×10^{-8}, regions in every event; `CC1mu1piXp` bitwise unchanged.
- 28 release samples and 18 detector-variation samples processed.
- Bin configuration: signal region as an ordinary bin, the four sidebands as sideband bins in the signal block (their signal is predicted).
- `Calculator Constrained` in the extraction configuration; the terms by source add up to the constrained total.

Detector-variation normalisation

Per POT the Run-4 detVar CV samples predict 1.27 (FHC) and 1.51 (RHC) times the nominal rate: they were simulated with the earlier NuMI flux. The absolute detector covariance carried that ratio.

- `DVfrac`: fractional change of each variation applied to the nominal prediction (also in the single-pion release since v1.8).
- Variation samples weighted to the current flux (per-event weight branch).

Constrained combined total: 17.9% (absolute) $\rightarrow$ 15.7% (fractional, with or without the flux weight).

Overtraining found: the two-fold event classifier

- Efficiency in the classifier training periods $14.04 \pm 0.21\%$, validation periods $13.16 \pm 0.31\%$ (2.4σ).
- Cause: the event classifier was trained on a random 80% of the training-period events. At score > 0.92, within the domain:

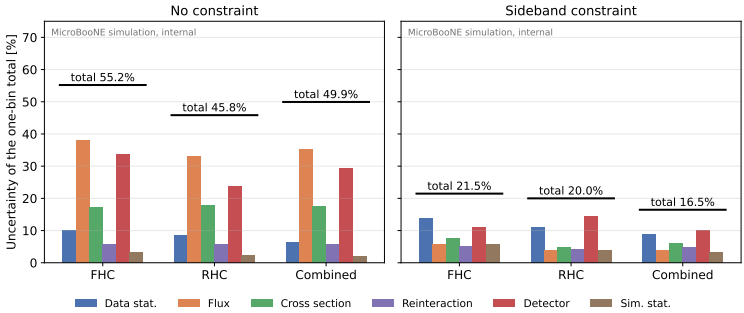
Events	Efficiency	Purity
its training events (80%)	$23.61 \pm 0.36\%$	55.4%
held-out 20%, same periods	$21.35 \pm 0.71\%$	51.0%
validation periods	$21.18 \pm 0.48\%$	51.4%

- On data: efficiency $\approx +6\%$, background $\approx -4\%$, about -2% on the total; invisible to closure and ensembles (their pseudo-data are the same events).

Fix: two folds by event number

- Models trained on even and on odd event numbers; every event, data included, scored by the model of the other parity. C++ = Python to 6×10^{-8}.
- Working point rescanned (study, constrained comb.): 17.4/20.4/16.5/19.5/25.5% at 0.85/0.88/0.90/0.92/0.94 $\Rightarrow$ 0.90.
- Signal region 920 events, 444 signal, purity 48%, efficiency 15%.
- All 46 samples reprocessed; total, variables and ensembles re-extracted (next slides).

One-bin total: uncertainties (two-fold classifier, working point 0.90)

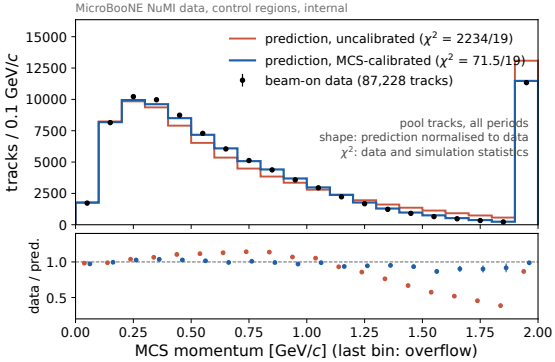


	Total	Data stat.	Detector	Unf./truth
FHC, constrained	21.5%	13.9%	10.9%	1.068
RHC, constrained	20.0%	11.0%	14.4%	1.059
Comb., none	49.9%	6.5%	29.3%	1.093
Comb., constrained	16.5%	8.9%	10.2%	1.059

- With the constraint, data statistics are the largest term.
- Single model: 15.2% (combined).
- MCS-calibration term, two-fold: 5.9/5.9/4.7% (FHC/RHC/comb.), 0.29 of the total.
- Unfolded/truth: one realisation; ensemble means within 1%.
- With it and the soft-pion term (next slides): 23.8/22.4/19.0% constrained.

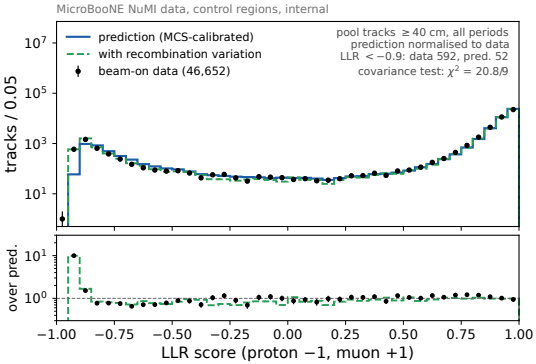
Beam-on data in the control regions: MCS momentum

- Inputs and output of the selection compared with beam-on data in the four opened control regions of the single-pion analysis; events of both signal regions removed. Shape tests per run period, summed.
- In the two-pion domain data/prediction 1.10–1.24, as outside it (1.12).
- MCS momentum of the pool tracks: data/prediction 1.14 at 0.7–0.9 GeV/c, 0.39 at 1.8–1.9 GeV/c, $\chi^2 = 2234/19$.
- The exiting-track effect of the single-pion analysis (angular noise at the MCS segment scale). Its v1.9 calibration map (cells of $|\cos\theta_x|$ and length) applied to the classifier inputs of every simulated exiting track: $\chi^2 = 71.5/19$; $P(\mu)$ 1100 $\rightarrow$ 293, $P(\pi)$ 256 $\rightarrow$ 56.

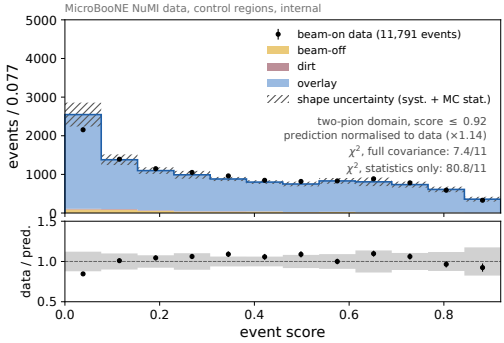


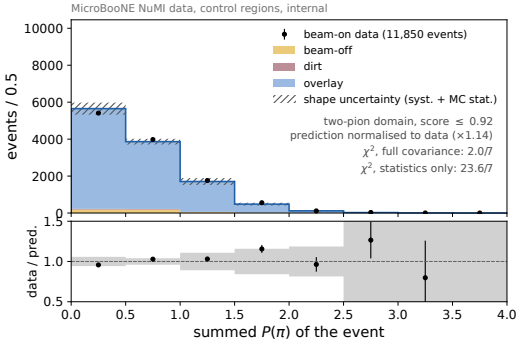
Beam-on data in the control regions: LLR of long protons

- Pool tracks longer than 40 cm with LLR < -0.9: $11\times$ more frequent in data (592 vs 52).
- Long contained protons: both ends contained (97%), from the vertex, 40–100 cm, true protons in 72% of the predicted; every period, both productions, all planes.
- Same hit count, more proton-like per hit in data: median LLR sum -436 to -486 (Run 4 data) vs -339 (simulation).
- Recomb2 reproduces it (median -445; share beyond -0.9: 0.15 vs 0.16–0.23 data, 0.015 CV); the wire-response variations move it the other way.
- The classifier still calls them protons: $P_p > 0.5$ for 86–92% in every LLR band, data and simulation within 1–2%.
- Recomb2 strength fitted with all other terms: 1.44 ± 0.50; results move by $\leq 0.40\sigma$, $\leq 0.12\sigma$ beyond the Recomb2 term.



Control regions: event score and summed $P(\pi)$





- Two-pion domain of the control regions (signal regions removed); prediction normalised to the data ($\times 1.14$); band: full systematic covariance of the shape.
- Event score with statistics only: $\chi^2 = 140/22$ after the calibration (160/22 before); the bin below the cut holds 0.83 of the predicted share, Recomb2 gives 0.86.

Control regions: shape tests with the systematic covariance

- Covariance of the prediction: flux, cross-section and reinteraction universes, the eight detector variations taken fractionally per horn mode (with the statistics of the variation samples), simulation and data statistics.
- Prediction normalised to the data total; the last non-empty bin dropped for the inversion, so ndf = non-empty bins -1; least-squares solve.
- Event score, class probabilities and MCS momentum are statistically compatible with the data under the full prediction covariance.
- The LLR score of the pool tracks remains incompatible with the data ($p = 0.014$) at its proton end, where the long protons sit; the detector term alone takes it from 398 to 36.

Quantity	Stat. only	Full covariance	p
Event score	98.5/11	6.5/11	0.84
Event score, two-fold	80.8/11	7.4/11	0.76
Summed $P(\pi)$	23.6/7	2.0/7	0.96
Track $P(\mu)$	48.1/9	4.2/9	0.90
Track $P(\pi)$	40.4/9	4.2/9	0.90
MCS momentum	65.2/8	6.2/8	0.63
LLR score	397.8/9	20.8/9	0.014

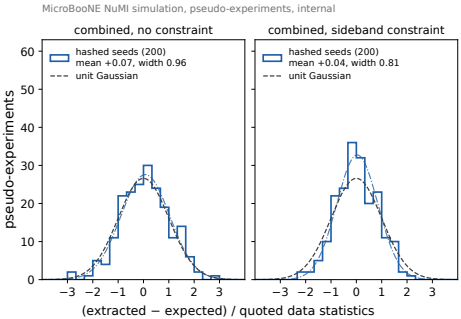
Same, coarser bins in both columns (the statistics-only values of the previous slides use finer bins).

MCS calibration in the extraction

- Overlays, dirt and detector-variation samples reprocessed with the calibrated classifier inputs; pseudo-data thrown with the same seeds (same events).
- Expected signal-region content changes by less than 1% (signal -0.9% FHC, -0.5% RHC); constrained combined total 15.7% $\rightarrow$ 15.2%.
- About 5% of the signal-region events migrate in or out; their pseudo-data counts are independent Poisson draws, which moves the single-realisation closure to 1.05–1.09 ($+0.9\sigma$ FHC, $+2.3\sigma$ RHC of the migrating events' Poisson spread).
- Calibration uncertainty as in the single-pion analysis: pseudo-data calibrated with each of the seven alternative maps, extracted with the nominal response, four fully correlated terms.
  - 2.2% of the constrained combined total, 2.3–5.5% elsewhere, at most 0.24 of the total (two-fold classifier: 4.7%, 0.29; shifts within $3.1\times$ their noise).
  - Each map moves 4–19 events; every shift within $1.7\times$ their Poisson noise: an upper estimate.
- The calibrated selection is used from here on.

Ensembles of pseudo-experiments

- Members thrown from the simulation (eight per-run files each), each extracted through the framework with and without the constraint; seed of each file a hash of (ensemble, member, file).
- Two-fold classifier, 200 members: offset $+0.0$ to $+1.0\%$ (± 0.5–0.8%), FHC, RHC, combined.
- Without the constraint: pull width 0.96–1.01, 70–72% inside the 68% interval, as expected.
- With the constraint: width 0.81–0.85 (76–78% inside): the quoted term propagates the data with the gain of the whole prediction (signal included), the estimator with the background gain: 8.6% vs 7.6% (ratio 0.88); with the fixed beam-off (0.98) the measured width.
- The first ensemble (100 members, consecutive seeds) had width 0.85 also without the constraint: an artefact of the seeding, gone with hashed seeds.



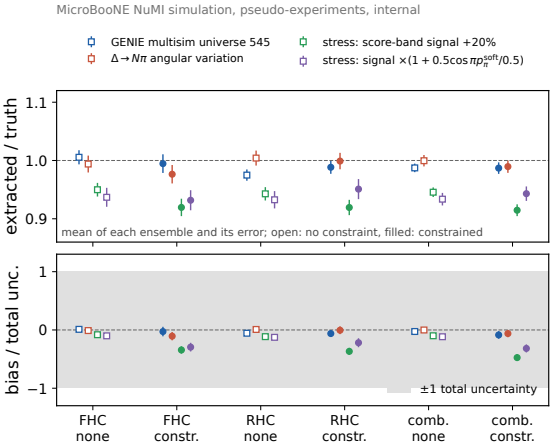
Pulls of the two-fold ensemble, combined configuration, without and with the constraint.

Ensembles under alternative and stress models

- 50 members each, nominal response; combined, constrained (single model → two-fold):

Model	single	two-fold
GENIE u545, $\Delta \to N\pi$	$\leq 0.17\sigma$	$\leq 0.11\sigma$
score-band signal +20%	-0.54σ	-0.47σ
$\geq 3\pi$ +30%, 1π −20%	$+0.04\sigma$	
signal $\times(1 + 0.5 \sin p_\pi^{\mathrm{lead}})$	-0.24σ	
signal $\times(1 + 0.5 \cos \pi p_\pi^{\mathrm{soft}}/0.5)$	-0.44σ	-0.32σ

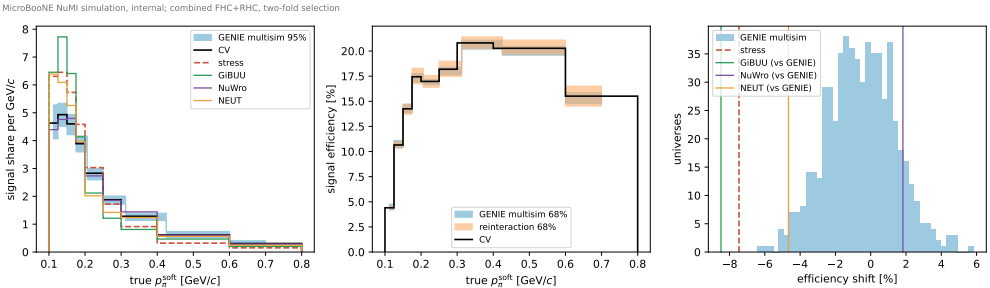
- Signal changes carried by the sidebands pull the constrained result towards the model; the soft-pion change is an efficiency effect (same bias without the constraint).
- Without the score band (framework, same members): total 20.4% (vs 15.9%), score-band stress -0.30σ, softest-pion unchanged.



Two-fold classifier: reweighted and stress models.

Soft pions: covariance against generators

- Efficiency 4% at $p_\pi^{\mathrm{soft}} = 0.10$–$0.125\,\mathrm{GeV}/c$, 14% at 0.15–0.175, 17–21% up to $0.6\,\mathrm{GeV}/c$, 15% above.
- 600 GENIE universes: 30–40% of the signal below $0.175\,\mathrm{GeV}/c$ (CV 35%); efficiency through the spectrum $\pm 1.2\%$. Stress model: 46%, -7.4% (3.8σ of the universes).
- Standalone generators, reweighted against GENIE: NuWro 35% / $+1.8\%$, NEUT 44% / -4.4%, GiBUU 52% / -8.3%. The stress model is inside the generator range, outside the covariance.
- A $0.175\,\mathrm{GeV}/c$ cut does not help (the softer events stay as background); migration across $0.10\,\mathrm{GeV}/c$ $< 1\%$.
- Added: soft-pion efficiency term = generator envelope, 8.5/8.1/8.3% (FHC/RHC/comb.), 4–12% per differential bin.



Selection variants

- Working point rescanned on the constrained combined total (score 0.85–0.96, study): 0.92 remains the minimum (16.8%; 15.2% through the framework). Tighter thresholds reach 58–70% purity at 22–32% total.
- Per horn mode: FHC minimum at 0.90 (25.9%; 27.3% at 0.92, within 2.7 points from 0.85 to 0.95); RHC at 0.92 (19.3%). One threshold kept.
- Shower features (energy, vertex distance, track score and direction of the leading shower; with dE/dx, second shower and two-shower mass in a larger set): ROC area 0.874 $\rightarrow$ 0.884–0.887, purity $+2$ points at fixed threshold.
  - In the π^0 control region the leading-shower energy, distance, score and direction follow the data; its dE/dx and the multi-shower features do not (χ^2 22–63 for 3–6 dof).
- Best constrained totals of the variants (with and without shower features, calorimetric particle classifier): 16.3–17.3% (21.1% calorimetric without showers) vs 16.8% adopted. The detector term of one variant moves by 3–6 points between neighbouring thresholds, more than these differences.
- The selection is kept.

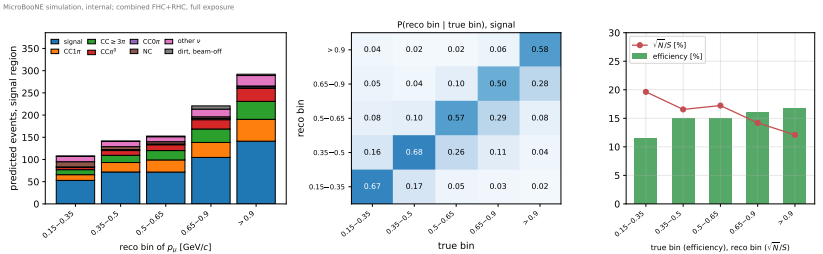
Measurements of variables: definitions and status

Variable	Reco (assigned tracks) / truth
p_μ	range if the end is contained, else MCS (v1.9 calibration for simulated tracks) / true p_μ
$\cos\theta_\mu$	about the target-to-vertex direction / about the true ν direction
$\cos\theta_\pi^{\mathrm{fwd}}$, $\cos\theta_\pi^{\mathrm{bwd}}$	the more forward and the more backward pion
$\cos\theta_{\pi\pi}$	opening angle of the two pions
$\cos\theta_{\mu\pi}$	muon and the closer pion
p_π^{low}, p_π^{high}	range momentum, both pions contained (pion mass from equal T/m)
N_p	unassigned track with $P_p > 0.5$ / proton above $0.3\,\mathrm{GeV}/c$

The pions are ordered by the variable itself, the same way at truth and reco level, so the convention is symmetric in the two pions.

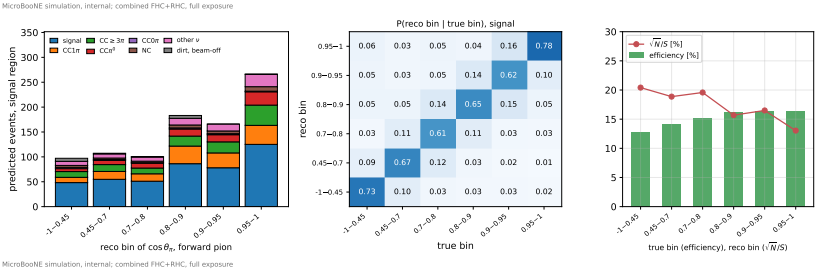
- Binning: contiguous bins with $P(\text{reco bin } i \,|\, \text{true bin } i) \geq 0.50$ (the criterion of the single-pion release) and at least 40 predicted signal events per bin.
- Combined, full exposure, two-fold classifier: signal region 920 events, 444 signal (all periods).
- $\sqrt{N}/S$: data statistics of the background-subtracted bin, before unfolding and constraint.
- Selection branches for these variables (equal to this study to 10^{-7}, all other branches unchanged); 46 samples reprocessed; univmake and the Wiener-SVD extraction per variable, with and without the constraint (results after the feasibility slides).

Muon momentum and angle



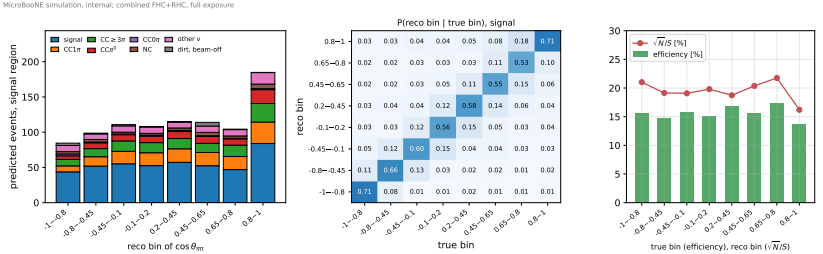
- p_μ: 5 bins, $P(r|t)$ 0.50–0.68, $P(t|r)$ 0.30–0.95, $\sqrt{N}/S$ 12–20%; MCS sets the resolution above 0.65 GeV/c.
- $\cos\theta_\mu$: 6 bins, $P(r|t)$ 0.73–0.91, 13–23%; forward-peaked, the last bin 0.95–1.
- Efficiency rises with p_μ and forward angle (10.5% → 15%).

Pion angles



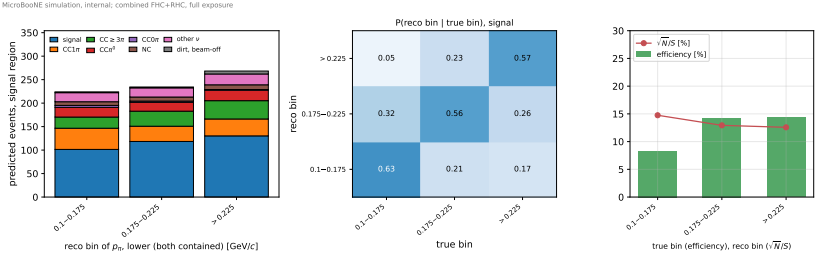
- Forward pion: 6 bins, $P(r|t)$ 0.61–0.78.
- Backward pion: 8 bins over the full range, $P(r|t)$ 0.64–0.83.
- $\sqrt{N}/S$ 13–22%; efficiency flat (11–15%).
- Angles are the best-measured pion quantities: the direction does not depend on the pion stopping.

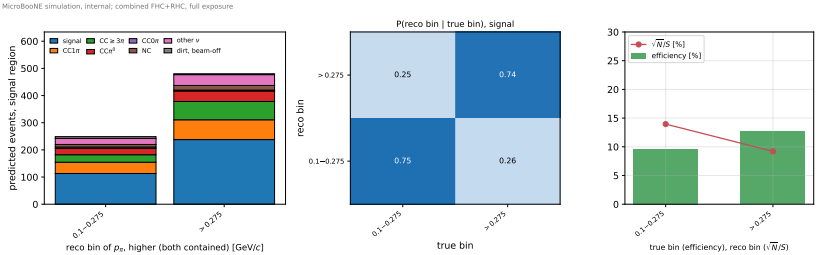
Opening angles



- $\cos\theta_{\pi\pi}$: 8 bins, $P(r|t)$ 0.53–0.71; sensitive to the resonance decay and to FSI.
- $\cos\theta_{\mu\pi}$ (closer pion): 7 bins, $P(r|t)$ 0.53–0.79.
- $\sqrt{N}/S$ 14–22% per bin.

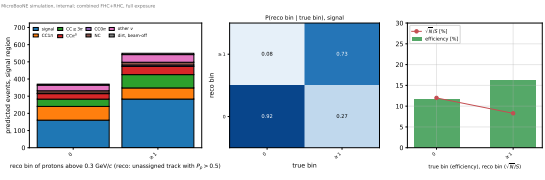
Pion momenta





- Both assigned pions contained in 79% of the selected signal, but the range momentum saturates (pions interact before stopping): only 3 (p_π^{low}) and 2 (p_π^{high}) bins pass.
- D2 allows pion-momentum bins only above $0.175\,\text{GeV}/c$: p_π^{low} would keep 2 bins.
- Low information; candidates for a coarse result only.

Proton multiplicity and summary

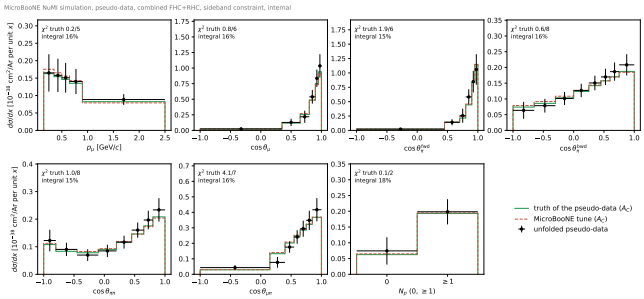


N_p: 0 and ≥ 1 protons above 0.3 GeV/c; $P(r|t)$ 0.92 and 0.73, $P(t|r)$ 0.36 and 0.98; $\sqrt{N}/S$ 12% and 8%. The $\geq 1p$ bin is the comparison with the BNB CC2$\pi^{\pm}Np$ result.

Variable	Bins	$P(r\|t)$	$P(t\|r)$	$\sqrt{N}/S$
p_μ	5	0.50–0.68	0.30–0.95	12–20%
$\cos\theta_\mu$	6	0.73–0.91	0.72–0.82	13–23%
$\cos\theta_\pi^{\mathrm{fwd}}$	6	0.61–0.78	0.56–0.82	13–20%
$\cos\theta_\pi^{\mathrm{bwd}}$	8	0.64–0.83	0.62–0.83	16–22%
$\cos\theta_{\pi\pi}$	8	0.53–0.71	0.53–0.82	16–22%
$\cos\theta_{\mu\pi}$	7	0.53–0.79	0.49–0.77	14–21%
p_π^{low}	3	0.56–0.63	0.35–0.84	13–15%
p_π^{high}	2	0.74–0.75	0.30–0.95	9–14%
N_p	2	0.73–0.92	0.36–0.98	8–12%

Two-fold classifier. $P(t|r)$: reconstructed-bin purity of the signal; lowest where neighbouring true bins feed in, corrected by the unfolding.

Differential extraction through the framework



Two-fold classifier, combined, sideband constraint: unfolded pseudo-data, A_C-smeared truth and tune.

Variable	Per bin none	Per bin constr.	χ^2 truth
p_μ	45–60%	17–32%	0.2/5
$\cos\theta_\mu$	40–123%	17–54%	0.8/6
$\cos\theta_\pi^{\mathrm{fwd}}$	51–113%	22–52%	1.9/6
$\cos\theta_\pi^{\mathrm{bwd}}$	45–99%	15–42%	0.6/8
$\cos\theta_{\pi\pi}$	43–66%	17–32%	1.0/8
$\cos\theta_{\mu\pi}$	40–185%	18–45%	4.1/7
N_p	44–130%	20–58%	0.1/2

- Wiener-SVD (filter on), FHC, RHC, combined; all 42 compatible with their truth in the nominal realisation.
- The constraint is essential: it halves the per-bin uncertainty or better.
- Integrals (full covariance) 15–18%, as the one-bin total; bin averages 21–39%.

Three or more pions

- In the two-pion domain, events with ≥ 4 tracks and $L_3 > L_2$: 234 at full exposure (combined, 43-input classifier), of which 88 have ≥ 3 charged pions, 41 a π^0, 35 are two-pion signal.
- A three-pion event classifier built the same way: ROC area 0.855, but every three-pion selection has a total uncertainty above 100% (`CC1mu3pi` 245%, with the particle classifier 117%, event classifier at best 105%).
- Four and five pions: 3.1 selected four-pion signal events at 4.8% purity and 0.1 five-pion.
- $P(\text{reco bin} \,|\, \text{true bin})$ of the pion multiplicity: 0.84, 0.48, 0.33 for 1, 2, ≥ 3 pions.

D8

The measurements stop at two pions. The three-pion-like events are a sideband of the two-pion constraint, as the $1\mu 3\pi$ sideband of the BNB note; alone it halves the total uncertainty.

To do

Before unblinding the one-bin total

- Decide between the nominal constraint (16.5%) and the reduced-transfer one without the score band (framework: 20.4%, half the signal-model transfer).
- Two-pion signal region opened only after the single-pion one (D7).

Towards the cross sections

- Generator predictions with the two-pion signal definition (GENIE tunes, NuWro, NEUT, GiBUU).
- Differential variables: extracted, compatible with their truth in the nominal pseudo-data; next ensembles and alternative models per variable, generator predictions. Invariant masses stay study products unless they pass the 0.50 criterion.
- The $\geq 1p$ subsample for the comparison with the BNB result.
- Independent-generator closure if a NuMI NuWro overlay can be validated (D5); NuGraph inputs if NuMI samples appear (D4).
- The same sideband constraint for the single-pion analysis, on pseudo-data.

Conclusions

- A $\nu_\mu + \bar{\nu}_\mu$ CC2$\pi^\pm$*Xp* selection on the full NuMI exposure (1.88×10^{21} POT), built on the two-stage strategy of the BNB CC2$\pi^\pm$*Np* analysis and adapted to NuMI: four-class particle classifier on LLR and geometry, best assignment, event classifier in two folds by event number (no overtraining). Combined: 15% efficiency, 48% purity, 920 predicted events (444 signal).
- Without a constraint the one-bin total carries 50%, dominated by the flux amplified by the background subtraction and by the detector variations. Four classifier-defined sidebands bring it to 16.5% combined (21.5% FHC, 20.0% RHC) through the framework; data statistics are then the largest term. With the MCS-calibration (4.7%) and soft-pion (8.3%) terms: 19.0%.
- Fixed for both analyses on the way: the detector-variation samples carry the earlier NuMI flux.
- Validation on pseudo-data: ensembles return the expected truth within 1%, with nominal coverage without the constraint and a conservative statistical term with it; alternative models close within 0.11σ; a score-band signal stress moves the constrained total by -0.47σ.
- Differential variables: p_μ, $\cos\theta_\mu$, the two pion angles, the opening angles and the proton multiplicity pass the 0.50 criterion with 2 to 8 bins at 8–23% data statistics per bin; the pion momenta give only 2 or 3 bins. Through the framework all seven are compatible with their truth in the nominal pseudo-data, 15–58% per bin (combined, constrained).
- Beam-on control regions: with the single-pion MCS calibration, the classifier inputs and the event score are compatible with the data under the full prediction covariance; the LLR score at its proton end (long protons) follows the recombination variation at 1.4 times its size ($\leq 0.12\sigma$ beyond the detector term).
- Three or more pions are not measurable at this exposure and serve as a control sample.

Backup

Backup: detector-variation treatments

	Absolute		Fractional		Fractional, flux-weighted	
	Detector	Total	Detector	Total	Detector	Total
FHC, no constraint	40.8%	58.4%	29.8%	51.3%	34.2%	54.0%
FHC, constrained	15.4%	24.9%	11.9%	22.4%	11.4%	22.5%
RHC, no constraint	28.4%	46.6%	17.9%	41.0%	18.1%	41.1%
RHC, constrained	11.1%	18.5%	7.6%	16.3%	7.3%	16.3%
Combined, no constraint	33.4%	50.7%	23.0%	44.5%	25.9%	46.0%
Combined, constrained	11.9%	17.9%	8.5%	15.7%	8.2%	15.7%

- Weighted to the current flux the detVar CV predicts 0.94 of the nominal rate, flat in energy; the origin of that offset is not identified, and the fractional treatment cancels it.
- The ratio variation/CV changes by at most 0.027 (median 0.003) with the flux weight; above 1 GeV it follows the integral within the variation-sample statistics (5–6%).
- Before the MCS calibration (the calibrated totals are on the uncertainty slide).

Backup: framework totals before and after the MCS calibration

		Total	Detector	Flux	Cross sect.	Data stat.	Unf./truth
FHC	none	54.0 → 51.3	34.2 → 30.9	37.0 → 36.2	14.7 → 14.6	10.5 → 10.4	1.056 → 1.082
FHC	constrained	22.5 → 21.3	11.4 → 10.6	6.1 → 6.2	8.7 → 7.8	14.0 → 13.5	1.020 → 1.050
RHC	none	41.1 → 40.3	18.1 → 19.3	31.5 → 30.1	15.5 → 15.0	8.7 → 8.5	1.035 → 1.090
RHC	constrained	16.3 → 15.4	7.3 → 6.6	4.9 → 4.2	5.9 → 5.6	11.1 → 10.7	1.020 → 1.070
Comb.	none	46.0 → 44.9	25.9 → 25.6	33.7 → 32.5	15.0 → 14.7	6.7 → 6.6	1.044 → 1.087
Comb.	constrained	15.7 → 15.2	8.2 → 8.0	4.8 → 4.6	6.6 → 6.3	8.8 → 8.7	1.021 → 1.056

Uncertainties in %. The unfolded/truth values are single pseudo-data realisations; the shift with the calibration is the Poisson noise of the migrating events.

Backup: the covariance-aware control-region test

- Prediction normalised to the data total in each test; the last kept bin dropped (the normalisation removes one degree of freedom); empty bins removed; χ^2 solved by least squares.
- Covariance: flux, cross-section and reinteraction universes of the CR skims; the eight detector variations taken fractionally per horn mode, with the statistics of the variation samples inside the detector term; simulation statistics; data statistics.

χ^2 with statistics and	flux	cross section	reinteraction	detector	full	data/pred.
Event score	49.8	11.2	58.3	53.0	6.5/11	1.139
Summed $P(\pi)$	18.3	6.8	11.5	3.1	2.0/7	1.139
Track $P(\mu)$	26.3	29.8	40.4	5.3	4.2/9	1.131
Track $P(\pi)$	17.8	17.5	22.7	6.8	4.2/9	1.131
MCS momentum	22.2	20.9	32.9	21.6	6.2/8	1.131
LLR score	382.1	364.1	319.7	36.0	20.8/9	1.131

Backup: working point per horn mode and the score band

- Constrained total vs threshold (study): FHC minimum 25.9% at 0.90, 27.3% at 0.92, within 2.7 points from 0.85 to 0.95; RHC minimum 19.3% at 0.92.
- The difference lies within the detector-term scatter between neighbouring thresholds; one threshold for both horn modes and the combination.
- Score-band to signal-region signal ratio: 1.684 (nominal), 1.660 (GENIE u545), 1.681 ($\Delta \to N\pi$); the two alternative models do not probe it.
- Stress ensembles: score-band signal $+20\%$ against the signal region (-0.54σ constrained, combined); $\geq 3\pi$ background $+30\%$ and 1π background -20% (within 0.18σ).